The reverse mathematics of the bounded Ramsey theorem for pairs

Quentin Le Houérou

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Joint work with Ludovic Patey

Roma Logica: The Forum

Bounded Ramsey theorem

Definition (BRT^2_2)

BRT^2_2 is the statement that:
"for every $\ell \in \mathbb{N}$ and every infinite graph $f : [\mathbb{N}]^2 \to 2$ with no clique of size ℓ, there exists an infinite anticlique."

BRT^2_2 is an immediate consequence of RT^2_2.

Proposition (Soldà (2021))

$\mathsf{RCA}_0 + \mathsf{EM} \vdash \mathsf{BRT}^2_2$

Proposition (Folklore (Belanger and Yokoyama for the proof over $\mathsf{RCA}_0 + \mathsf{I}\Sigma^0_2$))

BRT^2_2 *is computably true, meaning that for every instance* $f : [\mathbb{N}]^2 \to 2$*, there exists an* f*-computable solution.*

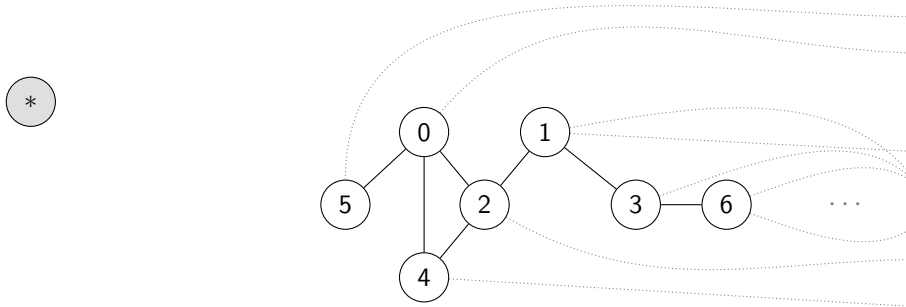
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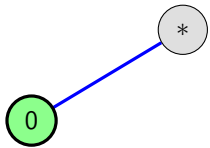
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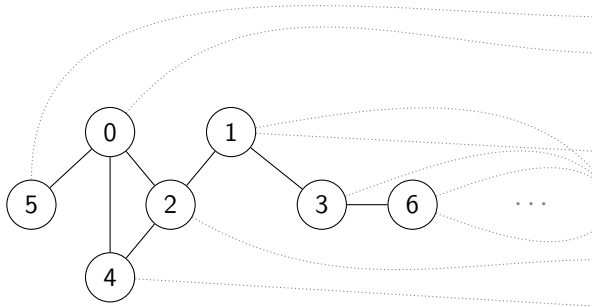
Proof (Belanger). Let $\ell \in \mathbb{N}$ and $f : [\mathbb{N}]^2 \to 2$ be an infinite graph with no clique of size ℓ. Consider the following f-computable tree:



Stage -1: the tree is reduced to its root $*$.

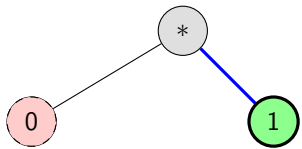
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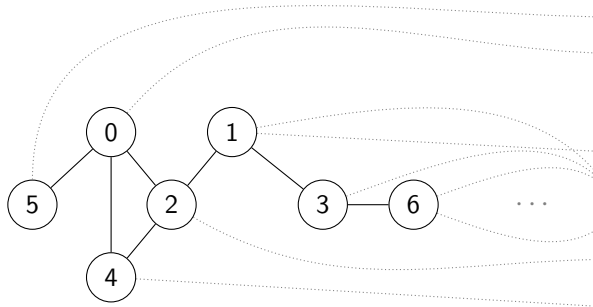




Stage 0: the root has no child, so 0 is added under $*$.

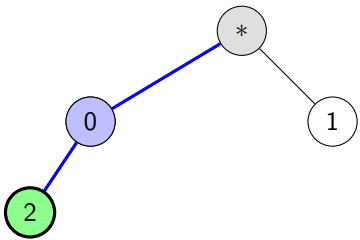
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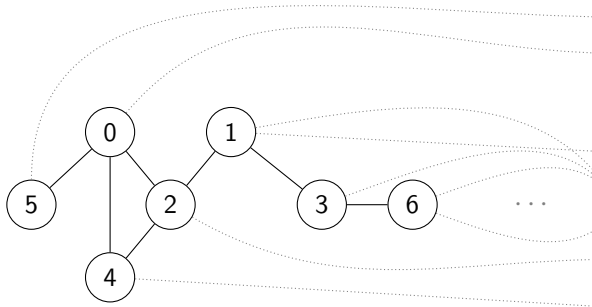




Stage 1: $1 \not\sim 0$, add 1 under $*$.

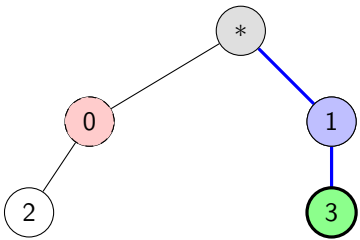
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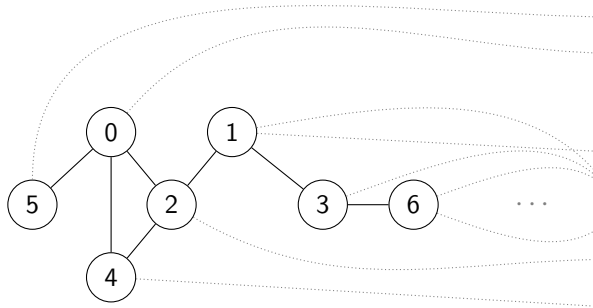




Stage 2: $2 \sim 0$, so we go down to 0 and 0 has no child, so 2 is added under 0.

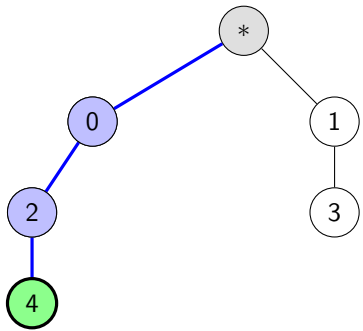
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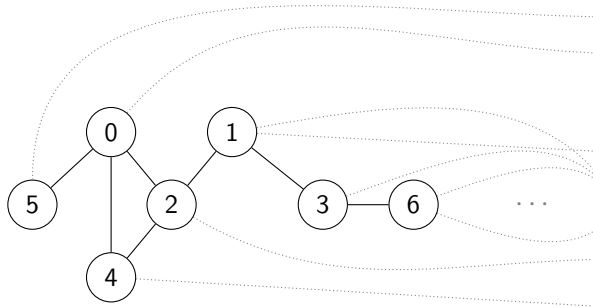




Stage 3: $3 \not\sim 0$ and $3 \sim 1$, so we go down to 1 and add 3 there.

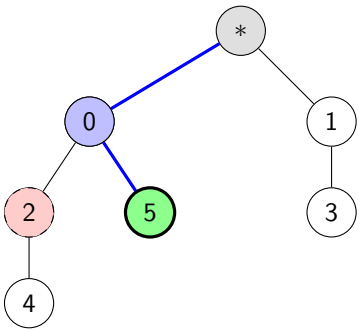
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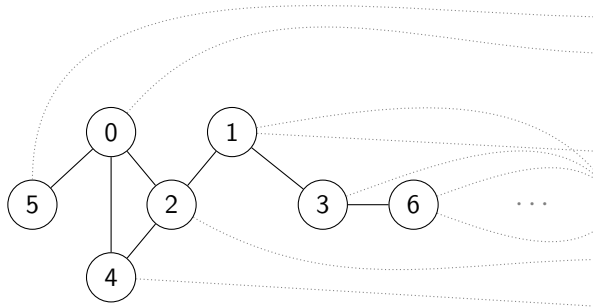




Stage 4: 4∼0 and 4∼2: we add 4 under 2.

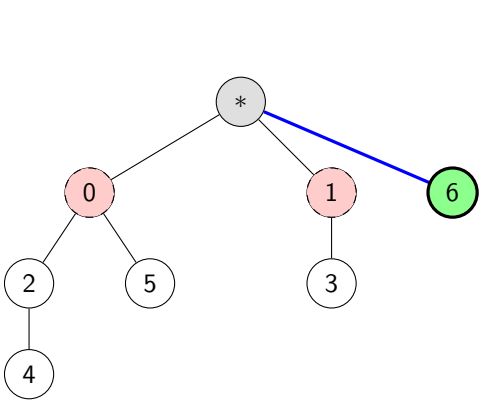
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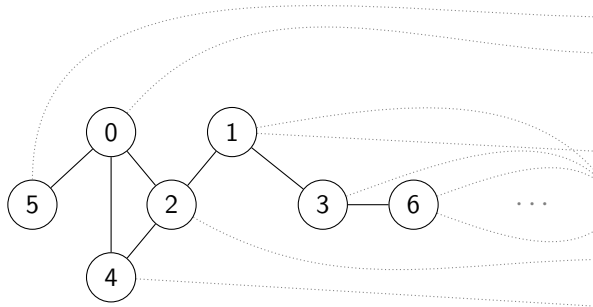




Stage 5: $5 \sim 0$, then $5 \not\sim 2$: we add 5 under 0.

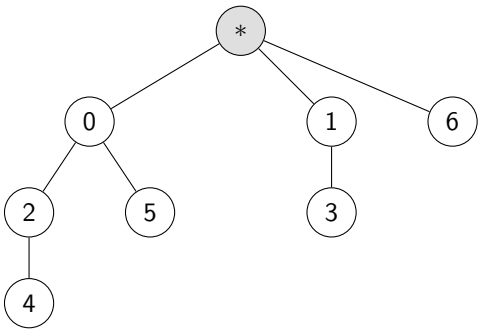
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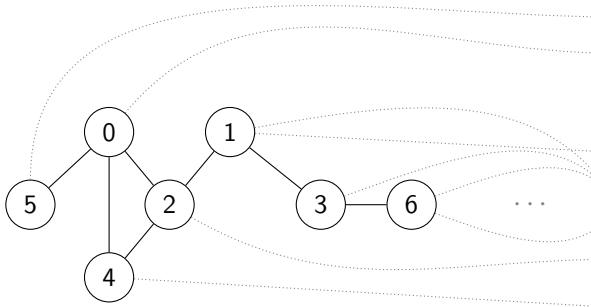




Stage 6: $6 \not\sim 0$ and $6 \not\sim 1$, so 6 is added under $*$.

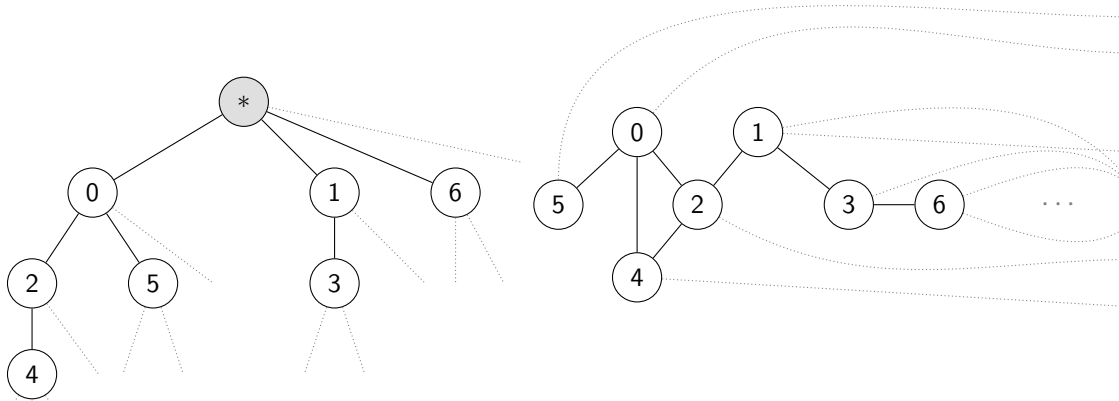
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Every path in the tree (without the root) forms a clique.
For every node, its direct descendants form an anticlique.

Proof (Belanger). Let $\ell \in \mathbb{N}$ and $f : [\mathbb{N}]^2 \to 2$ be an infinite graph with no clique of size ℓ. Consider the following f-computable tree:



After infinitely many steps, we obtain an f-computable tree of finite depth (bounded by $\ell + 1$). By $\mathsf{I}\Sigma^0_2$, there exists a node in that tree with infinitely many direct descendants, hence an infinite f-computable anticlique.

Proposition (Hirst)

$\mathsf{RCA}_0 + \mathsf{BRT}^2_2 \vdash \mathsf{B}\Sigma^0_2$ *and in particular* $\mathsf{RCA}_0 \nvdash \mathsf{BRT}^2_2$.

Theorem (L. and Patey)

The statement "BRT^2_2 *is computably true" is equivalent to* $\mathsf{I}\Sigma^0_2$ *over* RCA_0.

This may be surprising as RT^2_2 (and a fortiori BRT^2_2) is known not to imply $\mathsf{I}\Sigma^0_2$ over RCA_0 (Chong, Slaman, and Yang (2017)).
Thus, proving the existence of a computable solution is harder than merely proving the existence of one.

A similar phenomenon has been observed in Simpson and Yokoyama (2021) for a bounded width version of Kőnig's lemma. And more recently by Marcone and Volpi (2026) for theorems on the dimension of posets.

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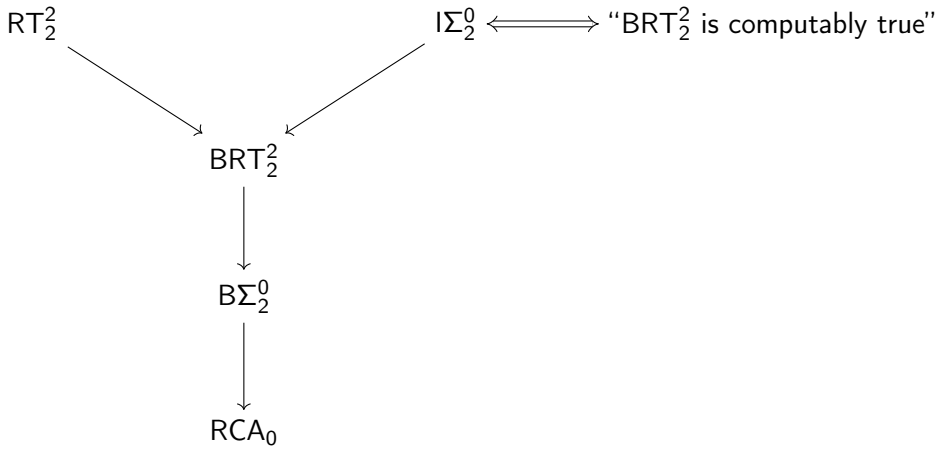
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Implications over RCA_0. A simple arrow represents a strict implication.

Avoiding a finite number of infinite sets

Lemma ($\mathsf{I}\Sigma_1$)

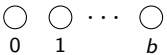
For every $b \in \mathbb{N}$ and $n \in \mathbb{N}$, there exists a computable graph $f_b^n : [\mathbb{N}]^2 \to 2$ such that:

- *f_b^n is triangle-free;*
- *If W_e is infinite for $e < n$, then $\exists x < y \in W_e \cap [b, +\infty)$ with $f_b^n(x, y) = 1$.*

f_b^n will be defined by stages, with $f_b^n(x, s)$ being defined at stage $s \in \mathbb{N}$ for every $x < s$.

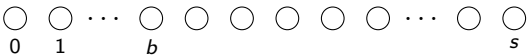
We say that $e < n$ is active at stage s if $\mathrm{card}(W_e[s] \cap [b, s]) > n^2$. Once an index is active, it stays that way.
By $\mathsf{I}\Sigma_1$, every $e < n$ such that W_e is infinite will eventually be active.

Construction of the graph



Until stage b, we do nothing

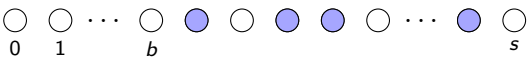
Construction of the graph



After stage b, we wait for some $e < n$ to become active.

Construction of the graph

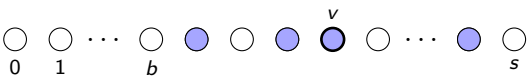
$|W_{e_0}[s] \cap [b, s]| > n^2$



If some $e_0 < n$ becomes active.

Construction of the graph

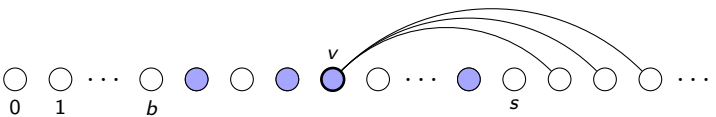
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If some $e_0 < n$ becomes active, we pick some element $v \in W_{e_0}[s] \cap [b, s]$.

Construction of the graph

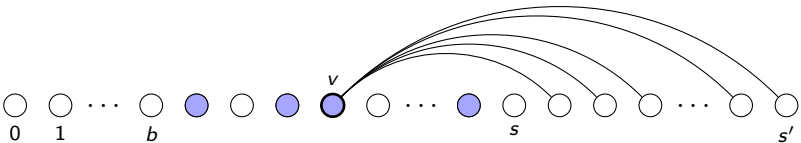
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In the following steps, add edges from v to the new vertices.

Construction of the graph

$|W_{e_0}[s] \cap [b, s]| > n^2$

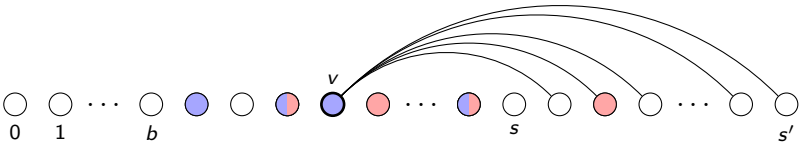


Continue like this until some new index becomes active.

Construction of the graph

$|W_{e_1}[s'] \cap [b, s']| > n^2$

$|W_{e_0}[s] \cap [b, s]| > n^2$

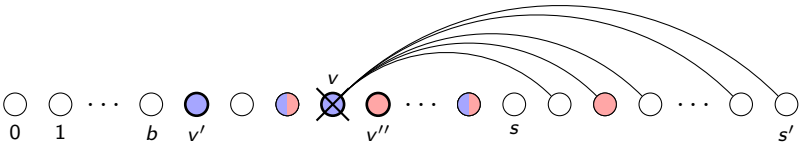


If some $e_1 < n$ becomes active.

Construction of the graph

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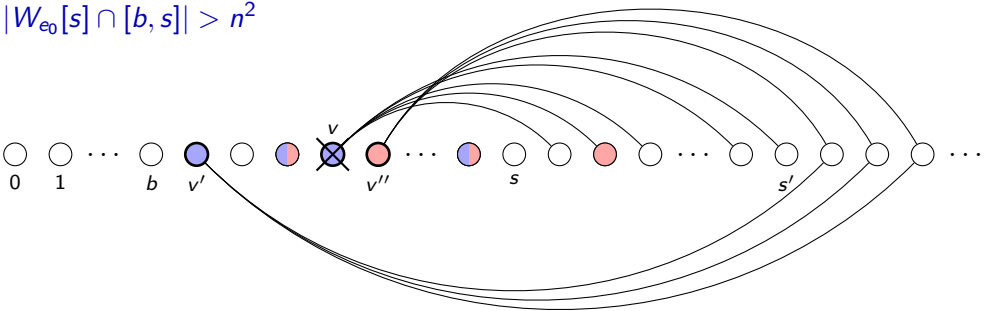


Put v in the trash and pick some $v' \in W_{e_0}[s] \cap [b, s]$
and some $v'' \in W_{e_1}[s'] \cap [b, s']$ not in the trash.

Construction of the graph

$|W_{e_1}[s'] \cap [b, s']| > n^2$

$|W_{e_0}[s] \cap [b, s]| > n^2$



The resulting graph will have no triangle (if $x < y < z$ forms a triangle, then x was picked at stage y but is then put in the trash in order for y to be picked at stage z).

If W_e is infinite, it becomes active at some stage s, and every element of W_e above s will be adjacent to one of the $> n^2$ in $W_e[s] \cap [b, s]$, hence W_e is not an anticlique.

Combining instances of BRT^2_2

Proposition ($\mathrm{I}\Sigma_1$)

Let $k \in \mathbb{N}$ and let $f^i : [\mathbb{N}]^2 \to 2$ be triangle-free for every $i < k$, then their union $f : [\mathbb{N}]^2 \to 2, (x, y) \mapsto \max_{i<k} f^i(x, y)$ has no clique of size ℓ for some $\ell \in \mathbb{N}$.

Proof.

$\mathrm{I}\Sigma_1$ proves the finite Ramsey theorem, hence let ℓ be such that, for every finite set E with $|E| \geq \ell$, every coloring $g : [E]^2 \to k$ admits a monochromatic triangle.

If f has a clique E of size ℓ, then we obtain a coloring $g : [E]^2 \to k$ by sending every pair $(x, y) \in [E]^2$ to one index $i < k$ such that $f^i(x, y) = 1$.
The assumption that the f^i are triangle-free contradicts the definition of ℓ. □

Avoiding all the computable sets

Theorem (L. and Patey)

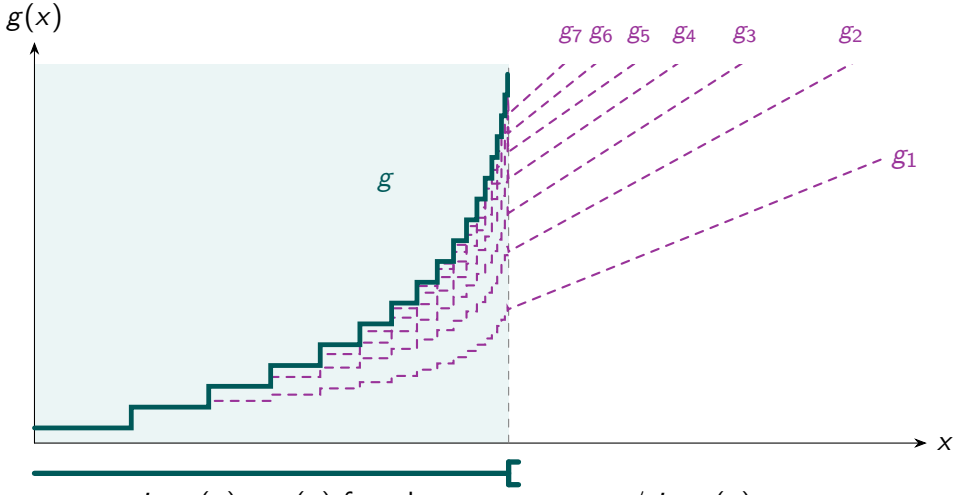
The statement "BRT^2_2 *is computably true" is equivalent to* $\mathsf{I}\Sigma^0_2$ *over* RCA_0.

- As $\mathsf{RCA}_0 + \mathsf{BRT}^2_2 \vdash \mathsf{B}\Sigma^0_2$, it is sufficient to show that every model of $\mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \neg\mathsf{I}\Sigma^0_2$ contains an instance $f : [\mathbb{N}]^2 \to 2$ of BRT^2_2 with no $\Delta_1(f)$ solution.
- For simplicity, we will work in models of $\mathsf{RCA}_0 + \mathsf{B}\Sigma_2 + \neg\mathsf{I}\Sigma_2$. Relativization is immediate.

When induction fails

Let $\mathcal{M} \models \mathrm{RCA}_0 + \mathrm{B}\Sigma_2 + \neg\mathrm{I}\Sigma_2$.

- there exists a Σ_2-formula $\theta(x)$ such that $\mathcal{M} \models \theta(0)$, $\mathcal{M} \models \forall x(\theta(x) \to \theta(x+1))$ but $\mathcal{M} \not\models \forall x\theta(x)$.
- By taking $\theta'(x) = \forall y \leq x\theta(y)$ we have that $\{x : \mathcal{M} \models \theta'(x)\}$ forms a cut I. By $\mathrm{B}\Sigma_2$, $\theta'(x)$ is equivalent to a Σ_2-formula $\exists y \forall z \varphi(x, y, z)$.
- Let $g_s : \mathbb{N} \to \mathbb{N}$ be defined by letting $g_s(x)$ be the smallest $y \leq s$ such that $\forall z \leq s\varphi(x, y, z)$ or be equal to $g_{s-1}(x)$ if no such y is found.
  1. $\lim_{s\to\infty} g_s(x)$ exists if and only if $x \in I$ and let $g : I \to \mathbb{N}$ be the limit function.
  2. $g.(\cdot)$ is non-decreasing in both parameters.
  3. g is non-decreasing and can be assumed to be increasing WLOG.
  4. By $\mathrm{B}\Sigma_2$, g takes unbounded values on I.



$x \in I$: $g_s(x) = g(x)$ for s large

$x \notin I$: $g_s(x) \to \infty$

Let $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma_2 + \neg\mathsf{I}\Sigma_2$ and let $I \subseteq M$, $g : I \to \mathbb{N}$ and $(g_s)_{s \in M}$ defined as before. Let $k > I$.
Let $f_b^n : [\mathbb{N}]^2 \to 2$ be the triangle-free graphs such that W_e is not an infinite anticlique when $e < n$.

For every $i < k$, define $f^i : [\mathbb{N}]^2 \to 2$ so that it is eventually equal to $f_b^{g(i)}$ for b the stage at which $g_b(i) = g(i)$ when $i \in I$.
Each f^i is triangle-free, and, for $i \in I$ and $e < g(i)$, W_e is not an infinite anticlique of f^i.

Then, $f : [\mathbb{N}]^2 \to 2, (x, y) \mapsto \max_{i<k} f^i(x, y)$ is a computable instance of BRT^2_2 with no computable solution.

Let $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma_2 + \neg\mathsf{I}\Sigma_2$ and let $I \subseteq M$, $g : I \to \mathbb{N}$ and $(g_s)_{s \in M}$ defined as before. Let $k > I$.
Let $f_b^n : [\mathbb{N}]^2 \to 2$ be the triangle-free graphs such that W_e is not an infinite anticlique when $e < n$.

For every $i < k$, define $f^i : [\mathbb{N}]^2 \to 2$ so that it is eventually equal to $f_b^{g(i)}$ for b the stage at which $g_b(i) = g(i)$ when $i \in I$.
Each f^i is triangle-free, and, for $i \in I$ and $e < g(i)$, W_e is not an infinite anticlique of f^i.

Then, $f : [\mathbb{N}]^2 \to 2, (x, y) \mapsto \max_{i<k} f^i(x, y)$ is a computable instance of BRT^2_2 with no computable solution.

Let $\mathcal{M} = (M, S) \models \mathrm{RCA}_0 + \mathrm{B}\Sigma_2 + \neg\mathrm{I}\Sigma_2$ and let $I \subseteq M$, $g : I \to \mathbb{N}$ and $(g_s)_{s \in M}$ defined as before. Let $k > I$.
Let $f_b^n : [\mathbb{N}]^2 \to 2$ be the triangle-free graphs such that W_e is not an infinite anticlique when $e < n$.

For every $i < k$, define $f^i : [\mathbb{N}]^2 \to 2$ so that it is eventually equal to $f_b^{g(i)}$ for b the stage at which $g_b(i) = g(i)$ when $i \in I$.
Each f^i is triangle-free, and, for $i \in I$ and $e < g(i)$, W_e is not an infinite anticlique of f^i.

Then, $f : [\mathbb{N}]^2 \to 2, (x, y) \mapsto \max_{i<k} f^i(x, y)$ is a computable instance of BRT_2^2 with no computable solution.

Other result

Theorem (L. and Patey)

$\mathsf{RCA}_0 + \mathsf{ADS} \not\vdash \mathsf{BRT}^2_2$.

The proof start from a model $\mathcal{M}$ of $\mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \neg\mathsf{I}\Sigma^0_2$ (containing an instance $f : [\mathbb{N}]^2 \to 2$ of BRT^2_2 with no solution), and extends it to a model N of $\mathsf{RCA}_0 + \mathsf{ADS} + \neg\mathsf{BRT}^2_2$ by iteratively adding solution to instances of ADS and avoiding adding solution to f.